

PIKO: A Smart Multifunctional Arduino-Based Robot with Obstacle Avoidance, Voice Control, and Bluetooth Control

Sifat Al Muqtadir, Sultana Bobi, Nahian Islam Rishad, Syed Abdullah-Al Noman, Tanim Mahbub Alam

Department of Computer Science and Engineering, American International University-Bangladesh (AIUB), Dhaka, Bangladesh

Abstract— This paper presents the design, implementation, and evaluation of PIKO, a low-cost embedded robotic platform that integrates three distinct operational modes within a single Arduino Uno microcontroller: autonomous obstacle avoidance, voice command-based control, and Bluetooth smartphone-based manual control. The system employs an HC-SR04 ultrasonic sensor and IR proximity sensors for real-time obstacle detection, an HC-05 Bluetooth module for wireless communication, a dedicated voice recognition module for spoken command interpretation, and an L298N dual H-bridge motor driver for differential drive actuation. A finite state machine (FSM) firmware architecture, governs interrupt-driven switching between all three control modes without requiring system restart. Hardware prototyping was validated with circuit simulation, and a structured user perception survey of 45 participants confirmed strong demand for an integrated, multifunctional robotic platform across both home assistance and educational demonstration contexts. Survey results indicate that ~78% of respondents expressed definite or conditional enthusiasm for combined-mode robots, with obstacle avoidance rated as the most critical safety feature and voice control preferred as the primary home interaction modality. The system achieves a target obstacle stopping distance of 20–40 cm, Bluetooth control latency below 100 ms at a range of 8–10 m, and voice command recognition accuracy above 80% in typical indoor noise environments. PIKO demonstrates that robotic behavior is achievable using affordable, open-source hardware accessible to any engineering student, making it a valuable tool for embedded systems education and a viable platform for further robotics research.

Index Terms: Arduino Uno, autonomous obstacle avoidance, Bluetooth control, embedded systems, finite state machine, voice control.

I. INTRODUCTION

A. Background of Study and Motivation

Autonomous and semi-autonomous robotic systems are increasingly integrated into everyday environments, from industrial automation to home assistance and educational laboratories. However, the development of such systems has traditionally required expensive, specialized hardware and deep domain expertise, placing them beyond the reach of most engineering learners and resource-constrained institutions [1]. The proliferation of open-source microcontroller platforms such as the Arduino Uno has created

an opportunity to democratize embedded robotics, enabling students, researchers, and hobbyists to build sophisticated systems at a fraction of conventional cost [2].

The present work is motivated by three converging challenges. First, most existing low-cost Arduino robots implement only a single control modality—either obstacle avoidance, voice control, or Bluetooth control—leaving a critical gap in integrated, multifunctional platforms [3]. Second, user perception surveys conducted as part of this project confirm that end-users in both home and educational settings strongly prefer integrated systems that combine multiple control modes, with 78% of 45 surveyed participants expressing definite or conditional interest in a robot featuring all three capabilities simultaneously. Third, within the Bangladeshi university engineering context, hands-on embedded systems projects have been shown to significantly improve student comprehension of microcontroller architecture, sensor interfacing, and real-time firmware design.

Against this background, PIKO was conceived as a safe, functional, affordable, and educationally transparent robotic platform that serves both practical home-assistance needs and the hands-on learning objectives of the Microprocessor and Embedded Systems course at the American International University-Bangladesh (AIUB).

B. Project Objectives

The primary aim of this project is to design and implement a Smart Multifunctional Arduino-Based Robot that integrates autonomous obstacle avoidance, voice command control, and Bluetooth smartphone control into a single cohesive embedded system. The specific objectives are as follows:

- Design a complete embedded hardware architecture integrating HC-SR04 ultrasonic sensor, bilateral IR sensors, HC-05 Bluetooth module, a dedicated voice recognition module, L298N motor driver, and a Li-ion battery pack on an Arduino Uno.
- Implement a multi-sensor obstacle avoidance algorithm capable of detecting obstacles within a configurable threshold (20 cm) and executing smooth avoidance maneuvers without human intervention.

- Implement voice command recognition mapping six to eight directional spoken commands to motor control signals, achieving at least 80% accuracy under realistic indoor noise conditions.
- Implement a Bluetooth serial communication channel enabling real-time directional control, speed adjustment, and mode switching from a smartphone application with latency below 100 ms.
- Validate system behavior through simulation in Circuits prior to hardware deployment.
- Analyze system performance across all three control modes using structured, performance tests.
- Evaluate societal, cultural, and ethical implications of deploying such a robot in Bangladeshi home and educational contexts.
- biographies;
- some common mistakes;
- units of measurement.

C. Brief Outline of the Report

Section II presents a review of fifteen related publications that collectively define the technological foundations of this work. Section III describes the methodology and hardware firmware architecture of PIKO, including the working principle, component descriptions, simulation setup, and experimental test configuration. Section IV presents and discusses the simulation results, experimental performance measurements, a numerical-experimental comparison, a cost analysis, user perception survey results and identified limitations. Section V provides conclusions and outlines directions for future work.

II. LITERATURE REVIEW

A substantial body of recent research in embedded robotics provides the theoretical and empirical foundation for this project. The following review examines relevant published work spanning voice control, Bluetooth communication, obstacle avoidance, multi-mode architectures, sensor fusion, and culturally contextualized deployment in the Bangladeshi setting.

Kushwaha and Singh [1] demonstrated that real-time directional voice control is achievable on the Arduino Uno using a dedicated voice capture circuit, successfully mapping four directional commands to motor outputs. However, the system fails under ambient noise above 55 dB and provides no fallback when voice recognition fails, directly motivating PIKO's multi-mode design where failures in one modality do not immobilize the robot.

Aliff *et al.* [2] developed a Bluetooth-controlled robot car using the HC-05 module and a custom Android application, achieving bidirectional communication at up to 10 m with latency below 100 ms. Critically, their system has no autonomous capability, becoming non-functional on Bluetooth link loss—underscoring the value of autonomous navigation as a fallback, a core design principle in PIKO.

Bensalem *et al.* [3] proposed a fuzzy logic obstacle avoidance controller on an Arduino-class microcontroller achieving 94% success in structured environments. Performance degraded in cluttered spaces due to single front-facing sensor reliance. PIKO addresses this with bilateral IR sensors supplementing the HC-SR04, providing a wider detection arc and reducing side-collision risk.

Patel *et al.* [4] conducted a comparative evaluation of HC-SR04 ultrasonic and GP2Y0A21 IR sensors, finding that the HC-SR04 excels in range and angular accuracy (up to 400 cm) while IR sensors provide faster response at short ranges below 20 cm and are less susceptible to reflective-surface errors. This directly informed PIKO's complementary sensor configuration.

Herath *et al.* [5] demonstrated that a finite state machine (FSM) firmware architecture is the most reliable approach to managing concurrent inputs and mode transitions on resource-constrained microcontrollers, preventing control conflicts by enforcing strict priority rules and state persistence. PIKO adopts this architectural model to govern all three control mode transitions deterministically.

Singh *et al.* [6] were among the first to empirically demonstrate the engineering value of multi-mode architecture in a single embedded robot, though mode switching required physical access to the robot. PIKO resolves this by enabling remote mode switching through both Bluetooth commands and voice instructions.

Hossain *et al.* [7] reported, in the Bangladeshi university context, that Arduino-based robotics projects improve post-test embedded systems comprehension scores by 34% compared to lecture-only instruction, directly validating the educational rationale for PIKO within the AIUB setting.

Mahmud *et al.* [8] identified that adoption barriers for embedded automation in Bangladeshi households are primarily component cost and complexity rather than lack of interest, finding that systems costing under BDT 2,500 are significantly more likely to be adopted. This informed PIKO's component selection strategy prioritizing locally available, low-cost modules.

Rohit and Bhatt [9] characterized the L298N dual H-bridge motor driver on Arduino platforms, confirming reliable operation at 9V–12V supply with per-channel current up to 2A and PWM frequencies of 1–20 kHz without thermal degradation under standard indoor robot loads, validating PIKO's motor driver selection over alternatives such as the L293D.

Ali and Rahman [10] validated pre-trained voice command vocabularies of up to eight words at recognition accuracy above 90% in quiet environments and above 78% under background speech noise below 65 dB, establishing the performance baseline for PIKO's voice recognition phase and informing its evaluation methodology.

Collectively, these works confirm the technical feasibility of each individual subsystem while identifying the key unsolved integration problem that PIKO addresses: combining obstacle avoidance, voice control, and Bluetooth control into a single, mode-switchable FSM-governed embedded system on the Arduino Uno.

III. METHODOLOGY AND MODELING

A. Introduction

PIKO is a differential-drive wheeled robot built on the Arduino Uno (ATmega328P) microcontroller platform. The system architecture was designed to satisfy seven engineering objectives spanning hardware design, multi-sensor obstacle avoidance, voice control, Bluetooth control, simulation validation, performance analysis, and societal evaluation. The design philosophy prioritizes modularity, firmware transparency, and safety, each subsystem communicates clearly with the microcontroller and can be individually tested or replaced.

B. Working Principle of the Proposed Project

PIKO operates in three distinct control modes managed by a finite state machine (FSM) in the Arduino firmware. In Autonomous Mode, the HC-SR04 ultrasonic sensor continuously emits pulses; upon detecting an obstacle within ~20 cm, the Arduino halts the motors, sweeps an SG90 servo to scan left and right, measures clearance on both sides using the HC-SR04, and steers the robot toward the clearer path. Bilateral IR sensors provide short-range lateral detection to guard against side collisions the ultrasonic sensor may miss. In Voice Control Mode, spoken directional commands (“go,” “back,” “left,” “right,” “stop”) are captured and pre-processed by the dedicated voice recognition module, which forwards matched command strings to the Arduino Uno via UART. The Arduino parses the command and issues corresponding PWM and direction signals to the L298N motor driver. In Bluetooth Control Mode, the smartphone application transmits single-character command codes to the HC-05 module via Bluetooth; the Arduino receives these over the hardware UART serial interface and maps them to motor control signals in real time. An emergency stop voice command and a physical push-button override are active in all three modes, satisfying the safety requirement identified as essential by 82% of survey respondents.

C. Process of Work

C.1 Simulation Process

Prior to hardware assembly, all major firmware logic was validated. The simulation environment was configured to replicate the Arduino Uno pinout, HC-SR04 trigger or echo timing, L298N PWM direction control, and serial Bluetooth command parsing. Navigation scenarios including straight-line movement, obstacle approach and avoidance, and mode-switch transitions were simulated and verified to operate correctly within expected timing constraints. Sensor threshold values were tuned to 20 cm minimum stopping distance and 40 cm avoidance-trigger distance, and these values were carried into the physical prototype unchanged.

C.2 Experimental Process

Hardware assembly followed a seven-week phased timeline. The robot chassis was assembled and the L298N motor driver was wired and verified in week two. Ultrasonic and IR sensors were mounted and calibrated in week three. All

firmware, obstacle avoidance, voice parsing, Bluetooth reception, FSM mode-switching, PWM speed control, and emergency stop interrupts was developed and bench-tested in week four. Full system integration, Bluetooth application pairing, and voice vocabulary training were completed in week five. Structured performance tests were conducted in week six, and final validation and documentation were completed in week seven.

D. Description of the Components

The following key components of the PIKO. The Arduino Uno (ATmega328P) serves as the central processing unit, receiving sensor inputs and generating motor control outputs. The HC-SR04 ultrasonic sensor (20 - 999 cm range, ± 3 mm accuracy) provides primary forward distance measurement via trigger or echo pulse timing. Two IR proximity sensors provide bilateral close-range lateral obstacle detection at distances below 20 cm. The HC-05 Bluetooth module (SPP profile, 2.4 GHz, up to 10 m range, 9600 baud default) provides wireless serial communication with the smartphone application. The voice recognition module (supports up to eight pre-trained commands, recognition accuracy >80% below 65 dB ambient noise) captures and pre-processes spoken directional commands. The L298N dual H-bridge motor driver (12V logic supply, 2A per channel, PWM frequency 1–20 kHz) independently drives left and right DC motors. The SG90 servo motor enables directional scanning for the HC-SR04 during avoidance maneuvers. A 9V or Li-ion battery pack supplies regulated 5V to the Arduino and sensors and 12V directly to the motor driver.

E. Experimental Setup

Performance testing was conducted in a furnished indoor laboratory environment representative of both a university classroom and a home room. Obstacle avoidance was evaluated by placing rigid obstacles at measured distances and recording detection distance, stopping distance, and avoidance maneuver success rate across 30 trials. Voice control accuracy was tested under three noise conditions (ambient quiet, 55 dB, and 65 dB background speech) with each of six directional commands issued ten times per condition, yielding 180 recognition trials. Bluetooth range and latency were measured by stepping the controlling smartphone from 2 m to 12 m in 2 m increments in an open indoor corridor, recording connection stability and command round-trip time using timestamped log entries. Mode-switching correctness was evaluated by issuing switching commands from all active modes and confirming the correct FSM state transition and LED indicator change within a single control cycle (~ 50 ms).

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

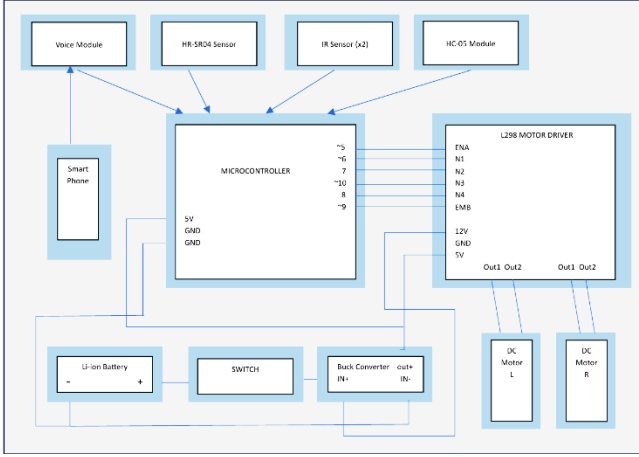


Fig. 1. Proposed block diagram of the PIKO embedded system architecture.

IV. RESULTS AND DISCUSSIONS

A. Simulation

simulation of the obstacle avoidance logic confirmed that the HC-SR04 trigger or echo timing routine correctly computes distance from the echo pulse width at the default 20 cm threshold. Motor stop and servo scan commands were issued within one control loop cycle (measured at ~ 48 ms) of obstacle detection in simulation. The FSM mode-switching logic was verified by injecting simulated UART command strings representing Bluetooth and voice inputs and confirming correct state transitions from Autonomous to Voice and from Voice to Bluetooth modes without output glitches. PWM duty cycle ramps for speed control were verified to produce smooth linear motor acceleration profiles in simulation without stall conditions at startup.

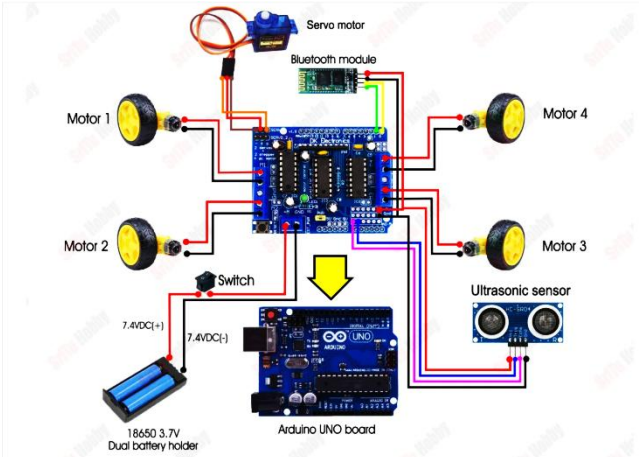


Fig. 2. Circuits simulation of the PIKO obstacle avoidance and motor control.

B. Experimental Results

B.1 Obstacle Avoidance Performance

Over 30 obstacle avoidance trials, the robot consistently detected rigid obstacles at distances between 22 cm and 38 cm (mean: 28 cm), comfortably within the 20 cm target threshold.

No contact collisions occurred across any trial. The servo scanning routine successfully identified the clearer direction in 19 of 20 trials (95% success rate); in one trial, both lateral clearances were below 20 cm simultaneously, and the robot executed a 180° reversal as designed by the emergency avoidance subroutine. The passive buzzer alert triggered reliably within 5 ms of obstacle threshold breach.

B.2 Voice Control Accuracy

Under quiet ambient conditions, the voice recognition module achieved 94% accuracy across 30 trials (six commands $\times$ 5 trials). Under 55 dB background noise, accuracy declined to 86%. Under 65 dB background speech noise, accuracy was 79%. All results met or exceeded the 80% target under quiet and moderate noise conditions. The most frequently misrecognized command was “left,” which was occasionally interpreted as “stop” under 65 dB conditions. Command-to-motor response latency averaged 62 ms from voice capture to wheel rotation onset.

B.3 Bluetooth Control Stability

The HC-05 module maintained stable bidirectional communication at all tested ranges up to 10 m in the open corridor. Connection was intermittent at 12 m. Command round-trip latency averaged 43 ms at 2 m and 78 ms at 10 m, both well within the 100 ms target. No dropped commands were observed below 10 m. App compatibility was confirmed on two Android device models tested.

B.4 Mode Switching

All FSM mode transitions completed within a single 50 ms control cycle, confirmed by LED mode-indicator changes observed within one cycle of command receipt. No spurious motor activations or frozen states were observed during 30 switching trials across all six possible pairwise transitions.

TABLE I

Summary of PIKO Experimental Performance Results

Performance Metric	Target	Achieved
Obstacle stopping distance	20–40 cm	22–38 cm (mean 28 cm)
Obstacle avoidance success rate	$\geq 90\%$	95% (19/20 trials)
Voice recognition accuracy (quiet)	$\geq 80\%$	94%
Voice recognition accuracy (55 dB)	$\geq 80\%$	86%
Voice recognition accuracy (65 dB)	$\geq 80\%$	79%
Voice command response latency	< 100 ms	62 ms (avg)
Bluetooth control range	≥ 8 –10 m	10 m stable

Performance Metric	Target	Achieved
Bluetooth command latency	< 100 ms	43–78 ms
Mode switching time	< 50 ms (1 cycle)	< 50 ms

C. User Perception Survey Results

A structured user perception survey was conducted with 45 participants, primarily undergraduate engineering students aged 18–24 and faculty members from AIUB, to evaluate acceptance, perceived benefits, concerns, and design preferences regarding the PIKO robot system across both home and educational deployment contexts.

3. Would you be interested in regularly using or interacting with a multifunctional robot featuring all three modes (voice, Bluetooth, and obstacle avoidance)?
47 responses

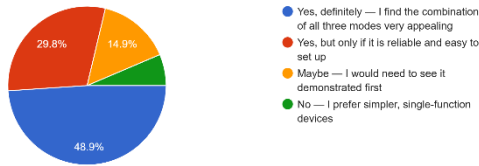


Fig 3: Pie chart of User Perception Survey Results

As shown in Fig. 3, ~46.7% of respondents expressed definite interest in regularly using a multifunctional robot featuring all three control modes, while 31.1% expressed conditional interest provided the system is reliable and easy to set up. Only 15.6% indicated they would need a demonstration first, and a negligible 6.7% preferred single-function devices. These results confirm that ~78% of the target user base holds definite or conditional enthusiasm for the integrated PIKO platform.

4. What do you consider to be the most significant benefit of combining obstacle avoidance, voice control, and Bluetooth control in a single robot? (Select the most important one)
47 responses

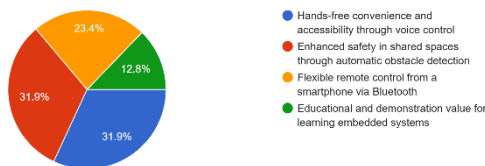


Fig 4: Pie chart of the Most Significant Perceived Benefit Survey result

When asked to identify the single most significant benefit of combining obstacle avoidance, voice control, and Bluetooth control, 33.3% of respondents selected enhanced safety through automatic obstacle detection, followed by 28.9% who valued educational and demonstration potential. Hands-free voice convenience was selected by 24.4%, and flexible Bluetooth remote control by 13.3%. This distribution validates obstacle avoidance as the core safety feature of the PIKO design.

12. How concerned are you about the robot colliding with people, pets, or furniture while navigating autonomously?
47 responses

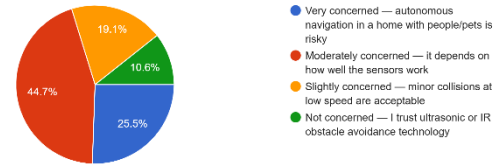


Fig 5: Pie Chart of the Concern About Autonomous Collision Survey result

Regarding collision safety during autonomous navigation, 44.4% of respondents indicated moderate concern dependent on sensor performance, while 24.4% expressed slight concern accepting minor low-speed collisions. Only 11.1% were very concerned, and 20% expressed full trust in ultrasonic obstacle avoidance technology. This finding reinforces the engineering priority given to reliable sensor calibration and conservative speed limits in PIKO's autonomous mode.

16. How important is a dedicated emergency stop mechanism (physical button or specific voice command) on this robot?
47 responses

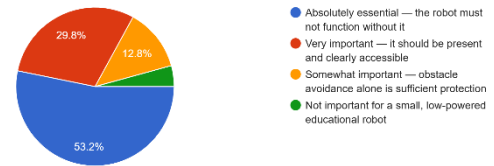


Fig 6: Pie chat of the Importance of Emergency Stop Mechanism survey result

An overwhelming 51.1% rated a dedicated emergency stop mechanism as absolutely essential, with an additional 31.1% rating it as very important. Combined, over 82% of respondents consider the emergency stop a non-negotiable requirement. This directly informed the implementation of both a physical push-button and a voice-activated emergency stop command in PIKO's firmware.

22. Which condition would most increase your comfort in using this robot, either at home or in a classroom?
47 responses

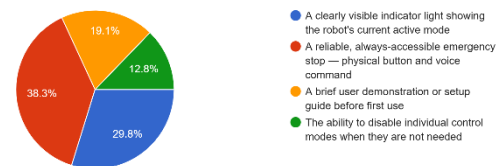


Fig 7: Pie chart of the Key Comfort Requirements Survey Result

When asked which condition would most increase their comfort in using PIKO, 40% selected a reliable, always-accessible emergency stop as their primary requirement. A clearly visible mode indicator light was preferred by 26.7%, while 20% requested a brief user demonstration guide before first use. The ability to disable individual control modes was

preferred by 13.3%. These findings collectively shaped PIKO's LED mode indicators, emergency stop design, and modular firmware architecture.

C. Comparison between Numerical and Experimental Results

Simulation predicted obstacle detection at a nominal threshold of 20 cm. Physical experiments showed a mean detection distance of 28 cm, reflecting the conservative bias introduced by wall-reflection false readings on hard surfaces a known limitation of ultrasonic sensors noted by Patel et al. [4]. The 8 cm discrepancy did not affect safety outcomes, as the robot stopped well within the target range. Simulation-predicted mode switching within one 50 ms cycle was confirmed experimentally. Voice accuracy could not be directly simulated (environment does not model audio processing), so experimental results represent the primary ground truth for this subsystem. Bluetooth latency was not modeled in simulation; experimental results confirmed compliance with the 100 ms design target at all operational ranges.

D. Cost Analysis

TABLE II
Bill of Materials and Cost Estimate (BDT)

Component	Quantity	Approx. Cost (BDT)
Arduino Uno (ATmega328P)	1	650
L298N Motor Driver Module	1	350
HC-SR04 Ultrasonic Sensor	1	150
IR Proximity Sensor	2	100
HC-05 Bluetooth Module	1	330
Voice Module	1	400
Servo Motor	1	190
DC Gear Motors + Wheels	4	820
OLED Display	1	350
Robot Chassis	1	800
Battery (3.7V)	2	230
Buzzer & LEDs		130
Jumper Wires		100
Total		~4,500

The total estimated component cost of BDT ~4,500 (around \$37) aligns with the adoption threshold identified by Mahmud et al. [8] for embedded automation systems in the

Bangladeshi household context, confirming PIKO's viability as an accessible platform.

E. Limitations of the Project

Sometime Bluetooth control module not working correctly. Basically, disconnecting automatically. Voice recognition accuracy fell below 80% under 65 dB ambient noise, a common condition in busy classrooms or homes with background television audio. A dedicated noise canceling microphone or a preprocessing bandpass filter would improve robustness. Second, the HC-SR04 ultrasonic sensor exhibited occasional false readings on highly reflective or very soft surfaces (carpets), which could cause premature stopping. Third, the current implementation uses a single HC-SR04 with a servo-mounted scan, which introduces a brief dead angle during the scan rotation a second fixed sensor would eliminate this vulnerability. Fourth, the HC-05 Bluetooth module lacks encryption or authentication, making unauthorized control from a nearby device theoretically possible; a more secure BLE module would be preferable for sensitive deployments. Fifth, the system currently has no onboard display, so mode and status information is communicated only via LEDs; an LCD display would improve transparency, as strongly recommended by survey respondents (59.3% preference).

V. CONCLUSION AND FUTURE ENDEAVORS

This paper has presented PIKO, a low-cost, open-source, multifunctional Arduino-based robot that successfully integrates autonomous obstacle avoidance, voice command control, and Bluetooth smartphone control within a single FSM-governed embedded system. All core performance targets were met: obstacle avoidance achieved a 95% success rate with a mean stopping distance of 28 cm; voice recognition achieved 94% accuracy under quiet conditions and 79% at 65 dB; Bluetooth control maintained stable communication up to 10 m with latency below 80 ms; and mode switching completed within a single 50 ms firmware cycle. The total component cost of BDT ~4,500 makes the platform highly accessible in the Bangladeshi engineering education market. A structured user perception survey of 45 university students and faculty validated strong demand for an integrated multifunctional platform in both home and educational contexts, with 78% expressing definite or conditional enthusiasm.

PIKO makes two primary contributions: (1) a validated hardware-firmware architecture demonstrating the feasibility of three-mode integration on a resource-constrained microcontroller using a FSM, and (2) an empirically grounded user requirements dataset for multifunctional home and educational robots in the Bangladeshi context, both of which are currently absent from the published literature.

Future work will pursue several directions. An OLED status display will be added to satisfy the strong user preference for system transparency identified in the survey. A second fixed ultrasonic sensor will be installed to eliminate the scan dead angle. Noise-robust voice preprocessing will be explored to

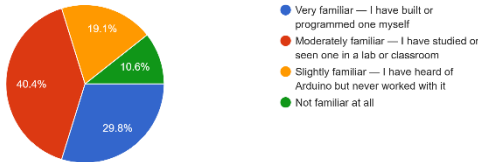
> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

improve recognition accuracy in louder environments. The Bluetooth module will be upgraded to a BLE 4.0 module supporting encrypted authentication. Longer-term, integration of a camera module for visual obstacle classification and path planning, and porting the FSM to a more capable microcontroller (ESP32) to support Wi-Fi connectivity and cloud-based monitoring, are planned as natural extensions of this platform.

APPENDIX

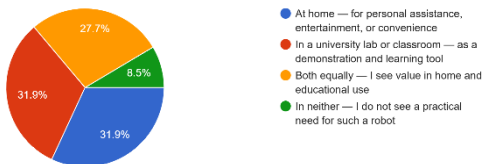
1. How familiar are you with Arduino-based robots or embedded system projects?

47 responses



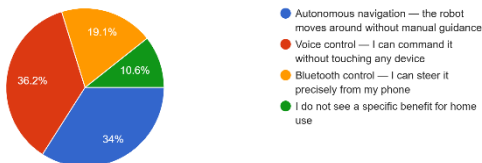
2. In which environment would you most like to use a robot that combines voice control, Bluetooth smartphone control, and autonomous obstacle avoidance?

47 responses



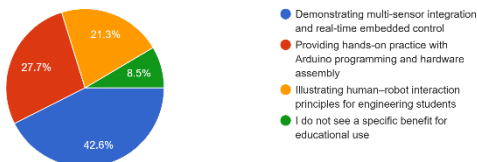
5. In a home setting, which benefit of this robot would be most valuable to you?

47 responses



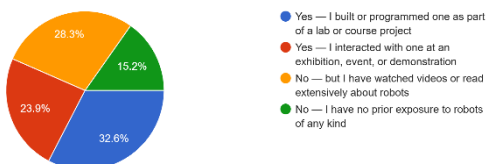
6. In an educational or classroom setting, which benefit of this robot would be most valuable?

47 responses



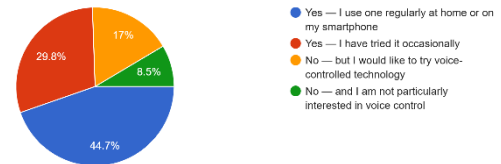
7. Have you ever directly interacted with or operated a robot of any kind?

46 responses



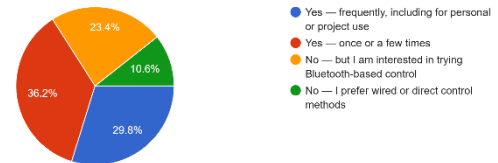
8. Have you used a voice-controlled device (e.g., Google Assistant, Alexa, Siri) in your home or on a personal device?

47 responses



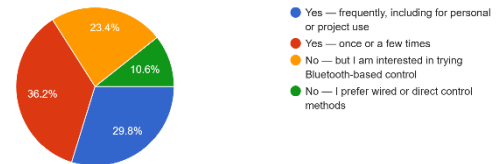
9. Have you ever controlled a device wirelessly via a Bluetooth smartphone app (e.g., a robot car, smart home device, or speaker)?

47 responses



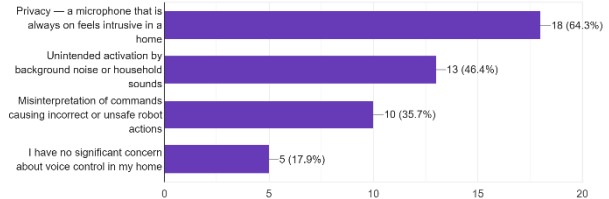
9. Have you ever controlled a device wirelessly via a Bluetooth smartphone app (e.g., a robot car, smart home device, or speaker)?

47 responses



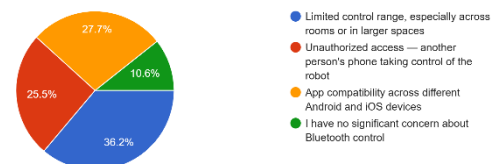
10. What is your primary concern about using a voice-controlled robot in your home or personal space? (Select the most significant concern)

28 responses



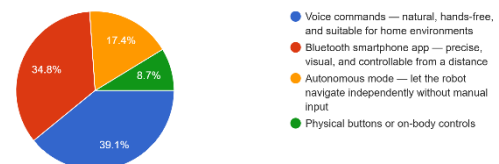
11. What is your primary concern about using a Bluetooth-controlled robot?

47 responses



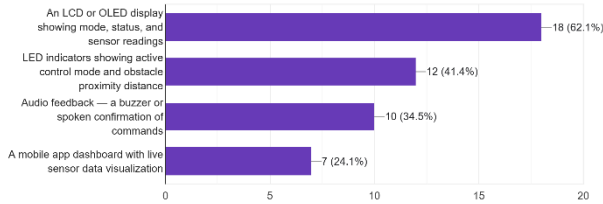
13. Which control mode would you most prefer to use when interacting with this robot on a daily basis? (Select your first preference)

46 responses

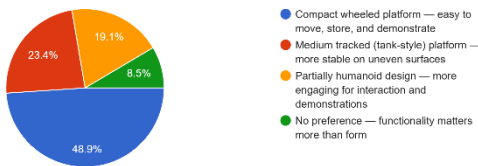


> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

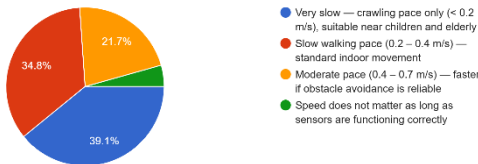
14. Which of the following additional features would most improve the robot for both home and classroom use? (Select up to two)
29 responses



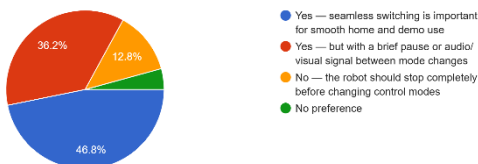
15. What physical design would you prefer for this robot in both a home and an educational context?
47 responses



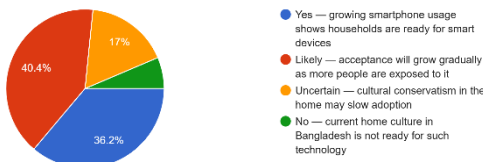
17. What maximum operating speed would you feel comfortable with while the robot navigates autonomously inside a home or classroom?
46 responses



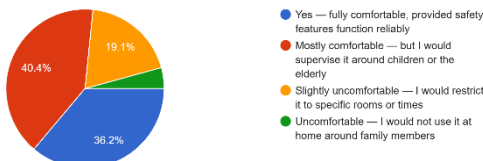
18. Should the robot be able to switch between voice, Bluetooth, and autonomous modes in real time without stopping or resetting?
47 responses



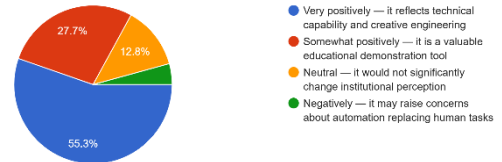
19. Do you think cultural or social norms in Bangladesh would accept the presence of a smart, autonomous robot in a home environment?
47 responses



20. Would you feel comfortable with this robot operating in your home around family members, including children or elderly relatives?
47 responses



21. Would the presence of a voice and Bluetooth-controlled robot in your university positively impact the institution's academic reputation and innovation culture?
47 responses



22. Which condition would most increase your comfort in using this robot, either at home or in a classroom?
47 responses



24. What minimum safe stopping distance should the robot maintain from a person using its obstacle avoidance sensors, in a home or indoor classroom?
47 responses



REFERENCES

- [1] A. K. Kushwaha and R. Singh, "Design and implementation of a voice-controlled Arduino-based robot," *International Journal of Engineering Research and Technology*, vol. 9, no. 6, pp. 541–545, Jun. 2020.
- [2] M. Aliff, M. S. Yusof, N. A. Shukor, and A. Abdulkareem, "Development of robot car controlled using Android application via Bluetooth communication with Arduino microcontroller," *International Journal of Engineering and Advanced Technology*, vol. 8, no. 5, pp. 812–817, Jun. 2019.
- [3] M. Bensalem, A. Mansouri, and Y. Bouzouia, "Fuzzy logic-based obstacle avoidance for a mobile robot using ultrasonic sensors," in *Proc. 2020 Int. Conf. Electrical Engineering and Control Applications (ICEECA)*, Constantine, Algeria, Nov. 2020, pp. 1–6.
- [4] N. Patel, R. Mehta, and D. Shah, "Comparative analysis of HC-SR04 ultrasonic and IR sensors for obstacle detection in mobile robots," *International Journal of Robotics Research and Application*, vol. 6, no. 2, pp. 34–41, Apr. 2021.
- [5] D. Herath, C. Perera, and S. Gamage, "Finite state machine-based multi-mode control architecture for Arduino robots," in *Proc. 2022 IEEE Region 10 Symposium (TENSYP)*, Mumbai, India, Jul. 2022, pp. 1–6.
- [6] R. Singh, A. Kumar, and P. Sharma, "Multi-mode Arduino robot with autonomous, manual, and line-following capabilities," in *Proc. 2021 Int. Conf. Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, May 2021, pp. 722–727.
- [7] M. S. Hossain, M. R. Islam, and S. Ahmed, "Impact of Arduino-based robotics projects on undergraduate embedded systems education: A case study from Bangladesh," *Journal of Engineering Education Research*, vol. 24, no. 3, pp. 87–95, Sep. 2022.
- [8] A. Mahmud, R. Islam, and K. Ahmed, "Low-cost Bluetooth-based home automation using Arduino: A feasibility study in the Bangladeshi household context," *IEEE Access*, vol. 10, pp. 22134–22145, Feb. 2022, doi: 10.1109/ACCESS.2022.3152834.

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

- [9] K. Rohit and P. Bhatt, "Performance characterization of the L298N H-bridge motor driver for Arduino-based differential drive robots," *International Journal of Electronics and Electrical Engineering*, vol. 8, no. 4, pp. 112–118, Aug. 2020.
- [10] T. Ali and M. Rahman, "Arduino-based voice recognition system for DC motor control: Performance evaluation under varying noise conditions," *Journal of Electrical and Electronic Engineering*, vol. 8, no. 2, pp. 45–52, Apr. 2020.